

TECHNICAL DOCUMENTATION — VenaFlow 3000

Device: VenaFlow 3000 | Class IIa active infusion pump | Non-IVD | Active | Software present

SECTION 1 — SAFETY AND INTENDED PURPOSE

The VenaFlow 3000 is an electronic infusion pump for controlled intravenous delivery of medication in hospital settings. A comprehensive risk management file has been prepared per ISO 14971:2019. Hazard analysis, FMEA, and risk controls are documented.

All residual risks have been assessed as acceptable given the clinical benefits.

The device has been designed and manufactured to ensure that when used by trained clinical staff for its intended purpose, risks to patients are minimised.

SECTION 2 — ELECTRICAL SAFETY AND SOFTWARE

Electrical safety testing has been conducted in accordance with IEC 60601-1:2018.

The device operates on mains power (240V AC) with a medical grade power supply.

Insulation, earthing, and fusing meet the requirements of IEC 60601-1. EMC testing was conducted per IEC 60601-1-2:2014+AMD1:2020. No electromagnetic interference was identified that would affect device function.

Software development followed IEC 62304:2006 medical software lifecycle process.

Software validation testing is documented in the software verification and validation report.

The software version is identified as VF3000-SW-v2.1.4 and is indicated on the device label.

SECTION 3 — MICROBIAL CONTAMINATION

The VenaFlow 3000 fluid path components are sterile as supplied. Sterilisation is by ethylene oxide per ISO 11135. Sterility is maintained until the expiry date under specified storage conditions. Biocompatibility of fluid-path materials was assessed per ISO 10993-1:2018 with biological evaluation studies conducted.

SECTION 4 — CLINICAL EVIDENCE

Clinical use data for the VenaFlow 3000 was compiled from peer-reviewed literature on equivalent infusion pump devices. Literature search conducted per MEDLINE, EMBASE (2010-2025). No serious adverse events attributable to device malfunction were identified. Clinical evaluation report (CER-VF3000-v1.0) is included in the technical documentation.

NOTE: No UDI documentation has been prepared. UDI compliance plan is pending.